QP CODE **Reg.No : .....**
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION MARCH 2026

FOURTH SEMESTER

Skill Enhancement Courses (SEC) – MG4SECMBG201 – Fundamentals of Microbiological Analysis of Food and Water

(2024 ADMISION ONWARDS)

Duration: 1 hr 30 min **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer **all** questions
Each question carries **1** mark

1. Identify the physical hazard from the following. [U] [1]
   a) Hair b) Pesrticides c) Bacteria d) Mycotoxins
2. Select the item which is not a part of personal protective equipment. [U] [2]
   a) Lab coat b) Mask c) Towel d) Gloves
3. HEPA filter is a part of which of the following. [K] [2]
   a) Autoclave b) Laminar air flow chamber
   c) Bacteriological incubator d) Hot air oven
4. Which of the following method is used for staining of fungi? [K] [3]
   a) Simple staining b) Lactophenol cotton blue staining
   c) Gram staining d) Differential staining
5. MacConkey agar is considered a selective medium because it: [A] [2]
   a) inhibits growth of Gram-negative bacteria
   b) inhibits growth of Gram-positive bacteria
   c) inhibits growth of Gram-negative and Gram-positive bacteria
   d) allows growth of Gram-negative and Gram-positive bacteria

6. Which of the following is used as the primary stain in Gram- staining? [K] [3]

a) Safranin b) Methylene blue c) Crystal violet d) Cotton blue

7. Indicate the media used for selective isolation of *Vibrio* species. [U] [4]

a) TCBC agar b) XLD agar

c) Baird Parker agar d) Sorbitol MacConkey agar

8. Select the method which is not a pure culture technique. [An] [5]

a) Streak plate b) Pour plate

c) Spread plate d) Serial dilution

9. Deoxycholate Citrate Agar is used for the isolation and differentiation of _____ from clinical specimens and food. [U] [4]

a) *Salmonella* and *Shigella* b) *Staphylococcus* and *E. coli*

c) *Salmonella* and *Staphylococcus* d) *Shigella* and *E. coli*

10. Select the temperature and time employed in autoclave sterilization. [K] [3]

a. 72 C, 15 sec b. 170 C, 1 hr c. 121 C, 15 min d. 100 C, 20 min

[1 x 10 = 10]

Part B

True or False questions

Answer **all** questions

Each question carries **1** mark

11. Poisonous mushroom is a type of chemical hazard. [U] [1]

12. The sample must contain lesser microbial load than the original food source. [U] [2]

13. Biosafety Level-3 laboratories deals with airborne pathogens that can cause serious or potentially lethal infections. [A] [2]

14. Coliforms are gram-negative rods that ferment lactose with acid and gas production. [U] [4]

15. MBRT is used to test the level of rancidity in milk. [U] [5]

[1 x 5 = 5]

Part C

One word or very short answer questions

Answer **all** questions

Each question carries **1** mark

16. What do you mean by aseptic condition? [U] [2]

17. Name two foodborne pathogens. [K] [1]

18. Indicate one purpose of swab testing. [A] [2]

19. Indicate the purpose of homogenization in sample prepararion. [U] [2]

20. What is rancidity? [K] [5]

21. Define coliforms. [K] [4]

22. Differentiate between sterilization and decontamination. [U] [3]

23. Indicate a test method to check the bacteriological quality of milk. [U] [5]

24. Expand FSSAI. [K] [2]

25. Name a culture medium used for the cultivation of fungi. [K] [3]

[1 x 10 = 10]

Part D

Reason based short answer questions

Answer any **5** questions

Each question carries **1** mark

26. Microorganisms in samples are considered biohazards. Give reasons. [U] [1]

27. *E. coli* gives red colour in Gram staining. Give reasons. [U] [3]

28. Food samples are serially diluted before pour plate culture method. Give reasons. [An] [5]

29. Food samples must be decontaminated before disposal. Explain the reason. [U] [3]

30. Give reasons why the bacteriological media are sterilized using an autoclave. [U] [3]

31. Explain why raw milk becomes colourless within 30 minutes during the MBRT. [A] [5]

32. Describe the biochemical basis for the formation of black-centred red colonies on XLD agar. [U] [4]

33. Basic stains are most suitable for staining bacterial cells. Give reasons. [U] [3]

[1 x 5 = 5]

Part E

Short answer questions

Answer any **5** questions

Each question carries **2** marks

34. Explain the method of simple staining. [U] [3]

35. Differentiate between physical and chemical hazards [U] [1]

36. Mention the composition and preparation of Nutrient agar. [U] [2]

37. Discuss about the significance of the FDA in public health. [U] [2]

38. Explain the method of estimation of acid value. [U] [5]

39. Describe the safe working practices followed inside a laminar airflow chamber. [U] [2]

40. Explain the method of fumigation and its significance. [U] [3]

41. Mention any two quality control practices that should be followed in a laboratory. [K] [2]

[2 x 5 = 10]

Part F

Long answer questions

Answer any **2** questions

Each question carries **5** marks

42. Explain the principle and working of a hot air oven. [U] [3]

43. Explain the method of estimation of peroxide value. [U] [5]

44. Explain the presumptive test of coliforms. [U] [4]

45. Write a short note on BIS. [U] [2]

[5 x 2 = 10]